Team ISEd Tea 001 Intrusion Report

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Overview:

Reporting on information acquired from 6:30 am to 9:00 am CST on Feb 21rd 2026.

WAF Activity Report:

Within each document section (for each box). Images from our IDS system will be used as a primary source of information. For consistency, please refer to these sections listed in the table of contents.

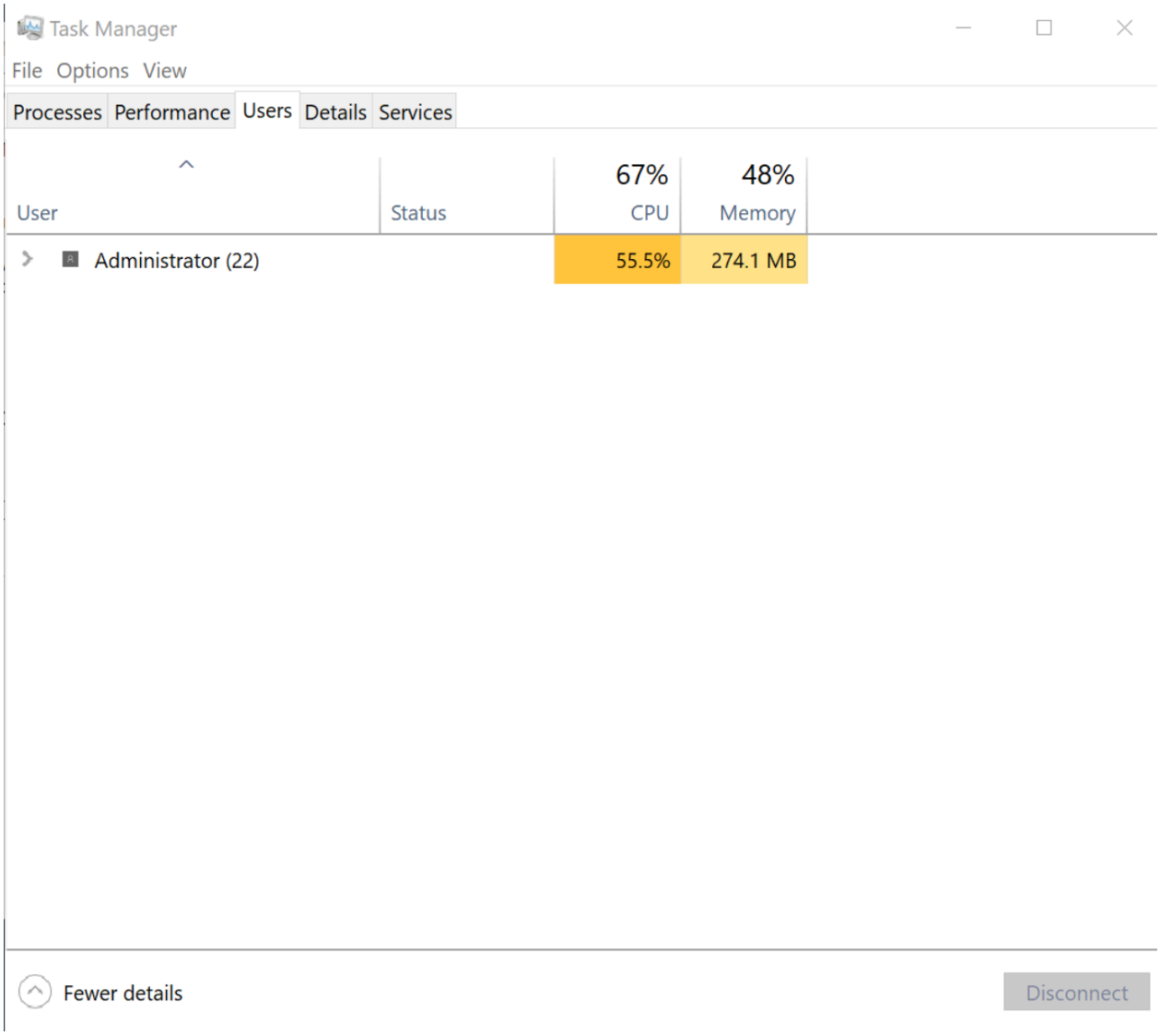
SEIM Activity Report:

Within each document section (for each box). Images from our IDS system will be used as a primary source of information. For consistency, please refer to these sections listed in the table of contents.

AD:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine

TICKETS:



We saw brute force attmepts for tickets but check

MGMT:

Bus and Trains Blue flag:
this issue started around 6:45 when we noticed failed login attempts via WUZAH and WAF, we then lost our flags at 7:04 for both our busses and trains

what we think happened:

There was a code vulnerability allowing anyone to hit the post request from the backend to turn on and off the busses regardless of their authentication status. The @require_auth decorator on the vehicle update endpoint contained a logic flaw where it returned a 401 status to the client but did not actually halt execution of the route function. This meant unauthenticated users could still modify vehicle status (enable/disable buses) despite receiving an 'unauthorized' response.

I verified this by sending POST requests without valid session cookies — the response was 401, but all vehicles were successfully enabled.

```
curl -s http://localhost:5000/vehicles/ | python3 -c "
import sys, json, subprocess
vehicles = json.load(sys.stdin)
for v in vehicles:
    vid = v['id']
    subprocess.run(['curl', '-b', 'cookies.txt', '-X', 'POST',
f'http://mgmt.team7.isucdc.com:5000/vehicles/%7Bvid%7D', '-H', 'Content-Type:
application/json', '-d', '{"status": true}'])
    print(f'Enabled {vid[:8]}')
"
```

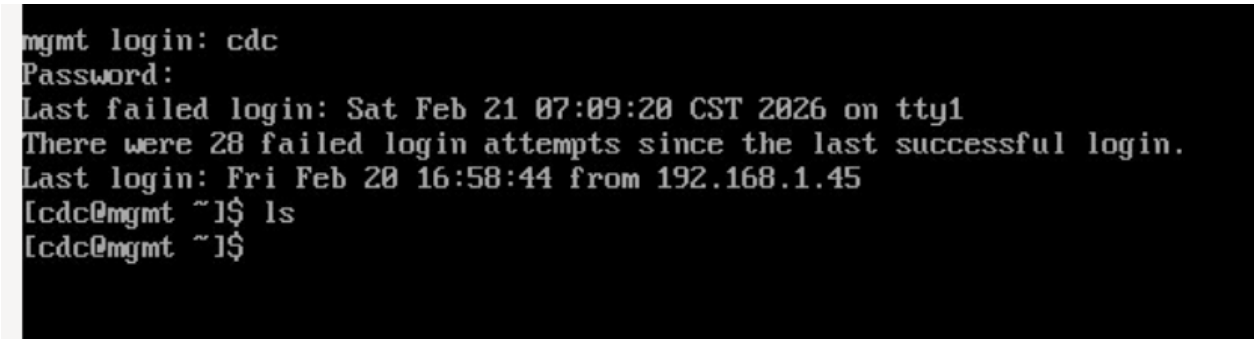
:thumbsup:
Click to react
:heart:
Click to react
:fire:
Click to react
Add Reaction
Reply
Forward
More
[8:30 AM]Saturday, February 21, 2026 8:30 AM
here is the output it returned:

```
{"error":"Authentication required"}
Enabled 0ab054ac
{"error":"Authentication required"}
Enabled 387cb109
{"error":"Authentication required"}
Enabled 3e409a45
{"error":"Authentication required"}
Enabled 6a53d7be
{"error":"Authentication required"}
Enabled 837b1b7e
```

{"error":"Authentication required"}
Enabled ae962416
{"error":"Authentication required"}
Enabled c5f60fcc
{"error":"Authentication required"}
Enabled c6de83f7
{"error":"Authentication required"}
Enabled c8177914
{"error":"Authentication required"}
Enabled ce83fa1f
{"error":"Authentication required"}
Enabled d351dac7
{"error":"Authentication required"}
Enabled e5ac8f47
{"error":"Authentication required"}
Enabled f3af1b4d
{"error":"Authentication required"}
Enabled ffac9a5f

We added logs so that we can see what is happening so we can develop a solution soon. It'll be easy to verify our solution and test it once we can see the logs

We plan to make it authenticated however when we apply auth to it, the users that are required to properly auth and control the buses cannot when we apply this change so we are in the process of development



Time	_source
Feb 21, 2026 @ 07:17:22.268	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:17:08 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 12.110.206.140 data.dstuser: cdc data.srcport: 56814 manager.name: wazuh2 rule.mail: false rule.level: 5 rule.hipaa: 164.312.b rule.pci_dss: 10.2.4, 10.2.5 rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication failed. rule.groups: syslog, sshd, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 13 rule.mitre.technique: Password Guessing, SSH rule.mitre.id: T1110.001, T1
Feb 21, 2026 @ 07:17:20.287	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:17:05 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.uid: 0 data.srcip: 12.110.206.140 data.euid: 0 data.dstuser: cdc data.tty: ssh manager.name: wazuh2 rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: PAM: User login failed. rule.groups: pam, syslog, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 22 rule.mitre.technique: Password Guessing
Feb 21, 2026 @ 07:17:20.285	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: unix_chkpwd predecoder.timestamp: Feb 21 13:17:05 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.srcuser: cdc rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: unix_chkpwd: Password check failed. rule.groups: pam, syslog, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 18 rule.mitre.technique: Password Guessing rule.mitre.id: T1110.001 rule.mitre.tactic: Credential Access rule.id:

Feb 21, 2026 @ 07:17:22.268	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:17:08 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 12.110.206.140 data.dstuser: cdc data.srcport: 56814 manager.name: wazuh2 rule.mail: false rule.level: 5 rule.hipaa: 164.312.b rule.pci_dss: 10.2.4, 10.2.5 rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication failed. rule.groups: syslog, sshd, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 13 rule.mitre.technique: Password Guessing, SSH rule.mitre.id: T1110.001, T1
Feb 21, 2026 @ 07:17:20.287	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:17:05 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.uid: 0 data.srcip: 12.110.206.140 data.euid: 0 data.dstuser: cdc data.tty: ssh manager.name: wazuh2 rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: PAM: User login failed. rule.groups: pam, syslog, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 22 rule.mitre.technique: Password Guessing
Feb 21, 2026 @ 07:17:20.285	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: unix_chkpwd predecoder.timestamp: Feb 21 13:17:05 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.srcuser: cdc rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: unix_chkpwd: Password check failed. rule.groups: pam, syslog, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 18 rule.mitre.technique: Password Guessing rule.mitre.id: T1110.001 rule.mitre.tactic: Credential Access rule.id:

We can see failed brute force attempts… these attempts started at 6:45am and ended at 6:48am

timestamp per 30 minutes

in session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id: 1771679106.14525989 full_log: Feb 21 13:04:51 mgmt.team7.isucdc.com ss

> Feb 21, 2026 @ 07:05:06.009

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:04:51 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.uid: 0 data.dstuser: roger rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Log in session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 10 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.tactic: Defense Evasion, Persistence, Privi

> Feb 21, 2026 @ 07:05:05.973

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: systemd predecoder.timestamp: Feb 21 13:04:51 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.uid: 0 data.dstuser: roger rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Log in session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 9 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.tactic: Defense Evasion, Persistence, Privil

> Feb 21, 2026 @ 07:05:05.921

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:04:50 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 68.32.254.50 data.dstuser: roger.cox data.srcport: 48956 manager.name: wazuh2 rule.mail: false rule.level: 3 rule.hipaa: 164.312.b rule.pci_dss: 10.2.5 rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication success. rule.groups: syslog, sshd, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 4 rule.mitre.technique: Valid Accounts, Remote Services rule.mitre.id: T1078, T1021

> Feb 21, 2026 @ 07:00:53.585

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:00:39 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 rule.firedtimes: 8 rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User authentication failure. rule.groups: syslog, access_control, authentication_failed rule.id: 2501 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8 rule.gdpr: IV_35.7.d, IV_32.2 location: journald decoder.name: sshd id: 1771678853.14377618 full_log: Feb 21 13:00

> Feb 21, 2026 @ 07:00:53.583

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:00:39 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 49.10.219.164 data.dstuser: games data.srcport: 53260 manager.name: wazuh2 rule.firedtimes: 13 rule.mail: false rule.level: 8 rule.description: Maximum authentication attempts exceeded. rule.groups: syslog, sshd, authentication_failed rule.mitre.technique: Brute Force rule.mitre.id: T1110 rule.mitre.tactic: Credential Access rule.id: 5758 rule.gpg13: 7.1 location: journald decoder.parent: sshd decoder.name: sshd

> Feb 21, 2026 @ 07:00:53.574

predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:00:39 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 rule.firedtimes: 7 rule.mail: false rule.level: 5 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User authentication failure. rule.groups: syslog, access_control, authentication_failed rule.id: 2501 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8 rule.gdpr: IV_35.7.d, IV_32.2 location: journald decoder.name: sshd id: 1771678853.14376687 full_log: Feb 21 13:00

Via our SIEM we can see login attempts … This log right here was when we attempted a login. We sshed from our internal machine so that we could use copy and paste to check the flags

```
[cdc@mgmt ~]$
[cdc@mgmt ~]$ curl -s http://localhost:5000/vehicles/ | python3 -c "
> import sys, json, subprocess
> vehicles = json.load(sys.stdin)
> for v in vehicles:
>     vid = v['id']
>     subprocess.run(['curl', '-b', 'cookies.txt', '-X', 'POST', f'http://localhost:5000/vehicles/{vid}', '
-H', 'Content-Type: application/json', '-d', '{\"status\": true}'])
>     print(f'Enabled {vid[:8]}')
> "
{"error":"Authentication required"}
Enabled 0ab054ac
{"error":"Authentication required"}
Enabled 387cb109
{"error":"Authentication required"}
Enabled 3e409a45
{"error":"Authentication required"}
Enabled 6a53d7be
{"error":"Authentication required"}
Enabled 837b1b7e
{"error":"Authentication required"}
Enabled ae962416
{"error":"Authentication required"}
Enabled c5f60fcc
{"error":"Authentication required"}
Enabled c6de83f7
{"error":"Authentication required"}
Enabled c8177914
{"error":"Authentication required"}
Enabled ce83fa1f
{"error":"Authentication required"}
Enabled d351dac7
{"error":"Authentication required"}
Enabled e5ac8f47
{"error":"Authentication required"}
Enabled f3af1b4d
{"error":"Authentication required"}
Enabled ffac9a5f
[cdc@mgmt ~]$ curl -s http://localhost:5000/vehicles/ | python3 -c "import sys,json; [print(f'{v[\"id\"][:8
]} status={v[\"status\"]}') for v in json.load(sys.stdin)]"
0ab054ac status=True
387cb109 status=True
3e409a45 status=True
6a53d7be status=True
837b1b7e status=True
ae962416 status=True
c5f60fcc status=True
c6de83f7 status=True
c8177914 status=True
ce83fa1f status=True
d351dac7 status=True
e5ac8f47 status=True
f3af1b4d status=True
ffac9a5f status=True
```

We then ran some commands and checked the code

```python
@bp.route("/<vehicle_id>", methods=["POST"])
@require_auth
def update_vehicle(vehicle_id):
    vehicle = Vehicle.query.get(vehicle_id)
    data = request.get_json()
    status = data.get("status")
    stop = data.get("current_stop")
    x = data.get("x")
    y = data.get("y")
    count = data.get("passenger_count")
    capacity = data.get("max_capacity")

    if not vehicle:
        return jsonify({"error": "Vehicle not found"}), 404

    if stop is not None:
        for i, s in enumerate(vehicle.route.stops):
            if s["name"] == stop:
                stop = i
                break
        else:
            return jsonify({"error": "No such stop"}), 404

        # flip direction for non-looping routes at end of route
        if not vehicle.route.looping and (
            (not vehicle.reverse and stop >= len(vehicle.route.stops) - 1)
            or (vehicle.reverse and stop == 0)
        ):
            print("REVERSE")
            vehicle.reverse = not vehicle.reverse

    vehicle.status = status if status is not None else vehicle.status
    vehicle.current_stop = stop if stop is not None else vehicle.current_stop
    vehicle.x = x if x is not None else vehicle.x
    vehicle.y = y if y is not None else vehicle.y
    vehicle.passenger_count = count if count is not None else vehicle.passenger_count
    vehicle.max_capacity = capacity if capacity is not None else vehicle.max_capacity

    try:
        db.session.commit()
        return jsonify({"message": "Vehicle updated successfully"}), 200
    except:
        return jsonify({"error": "Failed to update route"}), 500

```

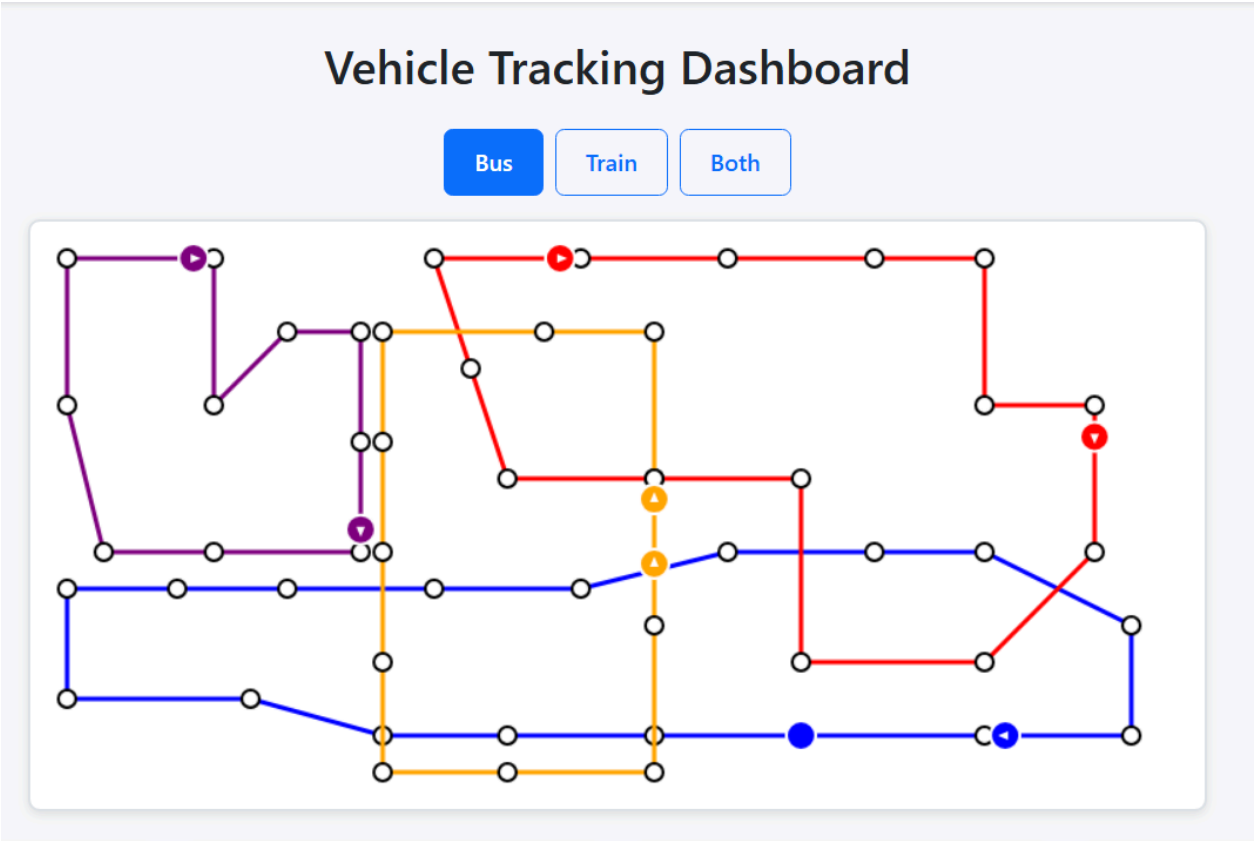
Here is the broken code… essentially if we were ISU cyride anyone, any student, anyone around the world could disable the bus and ticket system without any authentication because of the way @requireAuth works, it returns 401 but the request still goes through

U.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 19 rule.mitre.technique: Valid Accounts, Remote Services rule.mitre.id: T1078, T1021

Feb 21, 2026 @ 07:39:46.985 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:39:31 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 12.110.206.206 data.dstuser: root data.srcport: 52900 manager.name: wazuh2 rule.mail: false rule.level: 5 rule.hipaa: 164.312.b rule.pci_dss: 10.2.4, 10.2.5 rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication failed. rule.groups: syslog, sshd, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 17 rule.mitre.technique: Password Guessing, SSH

We also tried login into root with wireshark to check the pcap of the machine aswell.

ADMIN:



We saw that the buses went down and then we ran the code back to get the flags buses running again

```
12.110.177.57 - - [21/Feb/2026:07:24:54 -0600] "GET /auth/test HTTP/1.1" 401 36 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:147.0) Gecko/20100101 Firefox/147.0"

==> /var/log/modsec_audit.log <==
---5mtgnPDg---A--
[21/Feb/2026:07:24:54 -0600] 177168029456.723128 12.110.177.57 55099 192.168.1.35 5000
---5mtgnPDg---B--
GET /auth/test HTTP/1.1
Host: mgmt.team7.isucdc.com:5000
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:147.0) Gecko/20100101 Firefox/147.0
Cookie: session=eyJpc19hZG1pbiI6dHJ1ZSwidXNlcm5hbWUiOiJBZG1pbmlzdHJhdG9yIn0.aZmvVg.X9Hr8fOdjR8kkt6EcwJb9v4Pqe0
Accept-Encoding: gzip, deflate
Accept: */*
Connection: keep-alive

---5mtgnPDg---D--

---5mtgnPDg---F--
HTTP/1.1 401
Server: nginx/1.18.0
Date: Sat, 21 Feb 2026 13:24:54 GMT
Content-Length: 36
Content-Type: application/json
Access-Control-Allow-Origin: http://192.168.1.40
Connection: keep-alive
Vary: Origin

---5mtgnPDg---H--

---5mtgnPDg---I--

---5mtgnPDg---J--

---5mtgnPDg---Z--


==> /var/log/nginx/access.log <==
49.49.33.60 - - [21/Feb/2026:07:24:54 -0600] "GET /vehicles/ HTTP/1.1" 200 13800 "-" "Java-http-client/21.0.9"
199.100.16.100 - - [21/Feb/2026:07:24:54 -0600] "GET /api/vehicles HTTP/1.1" 200 13799 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Ch
rome/145.0.0.0 Safari/537.36 Edg/145.0.0.0"
49.49.33.60 - - [21/Feb/2026:07:24:55 -0600] "POST /vehicles/6a53d7be-5742-40d7-97ab-6215bed5d9b5 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
49.49.33.60 - - [21/Feb/2026:07:24:55 -0600] "POST /vehicles/c6de83f7-d539-4d40-905a-4fea924fc011 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
49.49.33.60 - - [21/Feb/2026:07:24:55 -0600] "POST /vehicles/c8177914-d1f0-4371-89fe-72a14453b664 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
49.49.33.60 - - [21/Feb/2026:07:24:55 -0600] "POST /vehicles/ffac9a5f-ecde-42e8-a3ec-e8f442b79009 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
49.49.33.60 - - [21/Feb/2026:07:24:55 -0600] "POST /vehicles/387cb109-7057-452a-ae73-b7518e947efc HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
199.100.16.100 - - [21/Feb/2026:07:24:55 -0600] "GET /api/vehicles HTTP/1.1" 200 13799 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Ch
```

We can see via our WAF failed 401 authentication attempts

These attempts started at 6:45am and ended at 6:48am

At 8:25am we thought we might have had a potential pam sniffer

```
cdc@mgmt pam.d]$ ls -la
otal 128
rwxr-xr-x.   2 root root    4096 Feb 11 16:58 .
rwxr-xr-x. 144 root root    8192 Feb 20 20:35 ..
rw-r--r--.   1 root root     272 Oct 10  2022 atd
rw-r--r--.   1 root root     192 Feb  4 14:18 chfn
rw-r--r--.   1 root root     192 Feb  4 14:18 chsh
rw-r--r--.   1 root root     910 Dec  1 02:40 cockpit
rw-r--r--.   1 root root     232 Dec 17 12:54 config-util
rw-r--r--.   1 root root     322 Apr  6  2024 crond
rwxrwxrwx.   1 root root      32 Feb 11 16:58 fingerprint-auth -> /etc/authselect/fingerprint-auth
rw-r--r--.   1 root root     715 Feb  4 14:18 login
rw-r--r--.   1 root root     154 Dec 17 12:54 other
rw-r--r--.   1 root root     168 Apr 18  2022 passwd
rwxrwxrwx.   1 root root      29 Feb 11 16:58 password-auth -> /etc/authselect/password-auth
rw-r--r--.   1 root root     510 Jul  9  2024 pluto
rw-r--r--.   1 root root     155 May 29  2024 polkit-1
rwxrwxrwx.   1 root root      25 Feb 11 16:58 postlogin -> /etc/authselect/postlogin
rw-r--r--.   1 root root     640 Feb  4 14:18 remote
rw-r--r--.   1 root root      97 Feb 19  2018 rhn_register
rw-r--r--.   1 root root     143 Feb  4 14:18 runuser
rw-r--r--.   1 root root     138 Feb  4 14:18 runuser-l
rw-r--r--.   1 root root     177 Nov  5 04:21 samba
rwxrwxrwx.   1 root root      30 Feb 11 16:58 smartcard-auth -> /etc/authselect/smartcard-auth
rw-r--r--.   1 root root     153 Dec 17 12:54 smartcard-auth.rpmnew
rwxrwxrwx.   1 root root      25 Dec  9 09:38 smtp -> /etc/alternatives/mta-pam
rw-r--r--.   1 root root      76 Oct 14  2023 smtp.postfix
rw-r--r--.   1 root root     727 Dec 17 20:27 sshd
rw-r--r--.   1 root root     214 Feb 10 11:01 sssd-shadowutils
rw-r--r--.   1 root root     566 Feb  4 14:18 su
rw-r--r--.   1 root root     154 Dec 18 08:01 sudo
rw-r--r--.   1 root root     178 Dec 18 08:01 sudo-i
rw-r--r--.   1 root root     137 Feb  4 14:18 su-l
rwxrwxrwx.   1 root root      27 Feb 11 16:58 system-auth -> /etc/authselect/system-auth
rw-r--r--.   1 root root     368 Jan 26 03:30 systemd-user
rw-r--r--.   1 root root      84 Oct 14  2023 vlock
rw-r--r--.   1 root root     159 Jan 23 02:36 vmtoolsd
rw-r-----.   1 root pegasus  277 Aug 31  2020 wbem
```

However after checking the pam directory

```
-rw-r--r--.   1 root root     159 Jan 23 02:36 vmtoolsd
-rw-r-----.   1 root pegasus  277 Aug 31  2020 wbem
[cdc@mgmt pam.d]$ ls -la
total 128
drwxr-xr-x.   2 root root    4096 Feb 11 16:58 .
drwxr-xr-x. 144 root root    8192 Feb 20 20:35 ..
-rw-r--r--.   1 root root     272 Oct 10  2022 atd
-rw-r--r--.   1 root root     192 Feb  4 14:18 chfn
-rw-r--r--.   1 root root     192 Feb  4 14:18 chsh
-rw-r--r--.   1 root root     910 Dec  1 02:40 cockpit
-rw-r--r--.   1 root root     232 Dec 17 12:54 config-util
-rw-r--r--.   1 root root     322 Apr  6  2024 crond
lrwxrwxrwx.   1 root root      32 Feb 11 16:58 fingerprint-auth -> /etc/authselect/fingerprint-auth
-rw-r--r--.   1 root root     715 Feb  4 14:18 login
-rw-r--r--.   1 root root     154 Dec 17 12:54 other
-rw-r--r--.   1 root root     168 Apr 18  2022 passwd
lrwxrwxrwx.   1 root root      29 Feb 11 16:58 password-auth -> /etc/authselect/password-auth
-rw-r--r--.   1 root root     510 Jul  9  2024 pluto
-rw-r--r--.   1 root root     155 May 29  2024 polkit-1
lrwxrwxrwx.   1 root root      25 Feb 11 16:58 postlogin -> /etc/authselect/postlogin
-rw-r--r--.   1 root root     640 Feb  4 14:18 remote
-rw-r--r--.   1 root root      97 Feb 19  2018 rhn_register
-rw-r--r--.   1 root root     143 Feb  4 14:18 runuser
-rw-r--r--.   1 root root     138 Feb  4 14:18 runuser-l
-rw-r--r--.   1 root root     177 Nov  5 04:21 samba
lrwxrwxrwx.   1 root root      30 Feb 11 16:58 smartcard-auth -> /etc/authselect/smartcard-auth
-rw-r--r--.   1 root root     153 Dec 17 12:54 smartcard-auth.rpmnew
lrwxrwxrwx.   1 root root      25 Dec  9 09:38 smtp -> /etc/alternatives/mta-pam
-rw-r--r--.   1 root root      76 Oct 14  2023 smtp.postfix
-rw-r--r--.   1 root root     727 Dec 17 20:27 sshd
-rw-r--r--.   1 root root     214 Feb 10 11:01 sssd-shadowutils
-rw-r--r--.   1 root root     566 Feb  4 14:18 su
-rw-r--r--.   1 root root     154 Dec 18 08:01 sudo
-rw-r--r--.   1 root root     178 Dec 18 08:01 sudo-i
-rw-r--r--.   1 root root     137 Feb  4 14:18 su-l
lrwxrwxrwx.   1 root root      27 Feb 11 16:58 system-auth -> /etc/authselect/system-auth
-rw-r--r--.   1 root root     368 Jan 26 03:30 systemd-user
-rw-r--r--.   1 root root      84 Oct 14  2023 vlock
-rw-r--r--.   1 root root     159 Jan 23 02:36 vmtoolsd
-rw-r-----.   1 root pegasus  277 Aug 31  2020 wbem
[cdc@mgmt pam.d]$ ls -la | grep  -i feb
drwxr-xr-x.   2 root root    4096 Feb 11 16:58 .
drwxr-xr-x. 144 root root    8192 Feb 20 20:35 ..
-rw-r--r--.   1 root root     192 Feb  4 14:18 chfn
-rw-r--r--.   1 root root     192 Feb  4 14:18 chsh
lrwxrwxrwx.   1 root root      32 Feb 11 16:58 fingerprint-auth -> /etc/authselect/fingerprint-auth
-rw-r--r--.   1 root root     715 Feb  4 14:18 login
lrwxrwxrwx.   1 root root      29 Feb 11 16:58 password-auth -> /etc/authselect/password-auth
lrwxrwxrwx.   1 root root      25 Feb 11 16:58 postlogin -> /etc/authselect/postlogin
-rw-r--r--.   1 root root     640 Feb  4 14:18 remote
-rw-r--r--.   1 root root      97 Feb 19  2018 rhn_register
-rw-r--r--.   1 root root     143 Feb  4 14:18 runuser
-rw-r--r--.   1 root root     138 Feb  4 14:18 runuser-l
lrwxrwxrwx.   1 root root      30 Feb 11 16:58 smartcard-auth -> /etc/authselect/smartcard-auth
-rw-r--r--.   1 root root     214 Feb 10 11:01 sssd-shadowutils
-rw-r--r--.   1 root root     566 Feb  4 14:18 su
-rw-r--r--.   1 root root     137 Feb  4 14:18 su-l
lrwxrwxrwx.   1 root root      27 Feb 11 16:58 system-auth -> /etc/authselect/system-auth
[cdc@mgmt pam.d]$ nvim rhn_register
-bash: nvim: command not found
[cdc@mgmt pam.d]$ vi !$
vi rhn_register
[cdc@mgmt pam.d]$ exit
logout
Connection to 192.168.1.30 closed.
cdc@Ubuntu-Internal:~$
```

We saw there was nothing on the box due to this because nothing was added today on the 21st nor is there any typically pam sniffer files.

WWW:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine